

Highlights

What limits reinforcement learning for distributed flutter suppression, and the credit structure that helps

- Physics-exact, closed-form credit signal for multi-agent flutter control.
- Per-agent control power, energy-normalised, verified additive to 10^{-10} .
- Credit signal plus log-energy shaping beats either alone ($d = -2.70$, $p = 0.002$).
- Exploration scale, not architecture or communication, gates visible gains.
- Matching exploration to actuation amplitude changes performance two-fold.